

Name:
Period:

Chapter 11 Review

11.1A/B Permutations and Combinations (2pts each)

1. The principal would like to assemble a committee of 7 students from the 15-member student council. How many different committees can be chosen?

$${}_{15}C_7 = 6435$$

2. A pizza shop has available toppings of olives, peppers, mushrooms, anchovies, pepperoni, bacon, and onions. How many different ways can a pizza be made with 3 toppings?

$${}_7C_3 = 35$$

3. In a running competition, a bronze, silver and gold medal must be given to the top three girls and top three boys. If 12 boys and 6 girls are competing, how many different ways could the six medals possibly be given out?

$${}_{12}P_3 \cdot {}_6P_3 = 158400$$

4. A committee must be formed with 5 teachers and 7 students. If there are 6 teachers to choose from, and 8 students, how many different ways could the committee be made?

$${}_6C_5 \cdot {}_8C_7 = 48$$

5. What is the total number of different 10-letter arrangements that can be formed using the letters in the word FACILITATE?

$$\frac{10!}{2! \cdot 2! \cdot 2!} = 453600$$

11.2 Probability (2pts each)

6. Which of these contexts describes a situation that is certain?

- ☒ a) Rolling a number less than or equal to 5 on a standard six-sided die, numbered from 1 to 6.
☒ b) Spinning a spinner divided into four equal-sized sections colored red/green/yellow/blue and landing on white.
☒ c) Winning a raffle that sold a total of 100 tickets if you bought 0 tickets.
☒ d) Reaching into a bag full of 10 strawberry chews and 10 cherry chews without looking and pulling out a strawberry or a cherry chew.

7. A spinner for a board-game is divided into four equal-sized sections colored red, green, yellow, and blue. If you land on a line between the colors, you keep spinning until you land on a color. Arianys' turn is next. Which word or phrase describes the probability that she will land on a primary color (red or yellow or blue)?

- a) certain
 b) impossible
☒ c) likely
 d) unlikely

8. Liam has a bag that contains pineapple chews, cherry chews, and lime chews. He performs an experiment. Liam randomly removes a chew from the bag, records the result, and returns the chew to the bag. Liam performs the experiment 62 times. The results are shown below:

- A pineapple chew was selected 58 times.
- A cherry chew was selected 2 times.
- A lime chew was selected 2 times.

Based on these results, express the experimental probability that the next chew Liam removes from the bag will be cherry chew as a decimal to the nearest hundredth.

$$\frac{2}{62} = 0.03$$

9. For a certain class, the following data table compares how many students have a brother or a sister. What is the probability that a randomly chosen student from the class has a brother?

	Has a brother	Does not have a brother
Has a sister	14	10
Does not have a sister	2	3

16

13

24

5

29

$$\frac{16}{29}$$

10. For a certain class, the following data table compares how many students have a cat or a dog. What is the probability that a student chosen randomly from the class does not have a cat?

	Has a cat	Does not have a cat
Has a dog	2	10
Does not have a dog	5	3

7

13

12

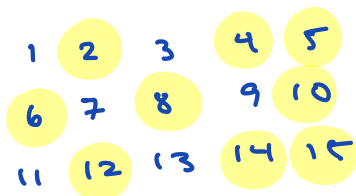
8

20

$$\frac{13}{20}$$

11.3 Probability of Multiple Events (2pts each)

11. There is a spinner with 15 equal areas, numbered 1 through 15. If the spinner is spun one time, what is the probability that the result is a multiple of 5 or a multiple of 2?



$$\frac{9}{15}$$

12. In a certain Algebra 2 class of 26 students, 18 of them play basketball and 17 of them play baseball. There are 5 students who play neither sport. What is the probability that a student chosen randomly from the class plays both basketball and baseball?

	Basketball	No Basketball	
Basketball	14	4	18
No Basketball	3	5	8
	17	9	26

$$\frac{14}{26}$$

13. For a certain class, the following data table summarizes the amount of sleep of students and whether or not they have an A in the class. What is the probability that a student chosen randomly from the class slept 6 hours or less and has an A?

	Slept more than 6 hours	Slept 6 hours or less
Has an A	3	2
Does not have an A	9	16

12

18

30

5

25

$$\frac{2}{30}$$

14. For a certain class, the following data table compares how many students have a cat or a dog. What is the probability that a student chosen randomly from the class has a cat ~~and~~ or a dog?

	Has a cat	Does not have a cat
Has a dog	5	2
Does not have a dog	12	4

17

6

23

7

16

$$\frac{19}{23}$$

15. In a class of students, the following data table summarizes how many students have a brother or a sister. What is the probability that a student has a sister given that they have a brother?

	Has a brother	Does not have a brother
Has a sister	5	9
Does not have a sister	3	13

8

22

30

14

16

$$\frac{5}{8}$$

11.9 Binomial Probability (2pts each)

16. If a fair coin is tossed 8 times, what is the probability, to the nearest thousandth, of getting exactly 2 tails?

$${}_8C_2 (0.5)^2 (0.5)^6 = 0.109$$

17. If a fair die is rolled 3 times, what is the probability, to the nearest thousandth, of getting exactly 1 three?

$${}_3C_1 \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^2 = 0.347$$

18. The Hiking Club plans to go camping in a State park where the probability of rain on any given day is $\frac{1}{4}$. What is the probability that it will rain on exactly one of the five days they are there? Round your answer to the nearest thousandth.

$${}_5C_1 \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^4 = 0.396$$

19. If a fair die is rolled 6 times, what is the probability, rounded to the nearest thousandth, of getting at least 5 fours?

$${}_6C_5 \left(\frac{1}{6}\right)^5 \left(\frac{5}{6}\right)^1 + {}_6C_6 \left(\frac{1}{6}\right)^6 \left(\frac{5}{6}\right)^0 = 0.001$$

20. If the probability that the Islanders will beat the Rangers in a game is 0.33, what is the probability that the Islanders will win at most two out of five games in a series against the Rangers? Round your answer to the nearest thousandth.

$${}_5C_0 (0.33)^0 (0.67)^5 + {}_5C_1 (0.33)^1 (0.67)^4 + {}_5C_2 (0.33)^2 (0.67)^3 = 0.795$$